

Expansion and Diversion in Global Green Markets: Evidence from the 2024 U.S. Election

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Abstract

Green industrial policies simultaneously broaden clean energy markets and shift competitive advantage toward domestic firms, making their global effects theoretically ambiguous. We estimate an event study using Donald Trump's 2024 election victory as a negative shock to expectations about the durability of the U.S. Inflation Reduction Act. We find no evidence of a negative aggregate effect consistent with a global expansion of clean energy markets. Instead, U.S. and Chinese firms exhibit large and opposite abnormal returns, indicating that investors expected the IRA's redistributive effects to dominate.

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Climate policy is widely implemented through instruments of “green industrial policy” (Rodrik 2014; Nemet 2019; Allan, Lewis and Oatley 2021; Allan and Nahm 2025). Across major economies, governments have adopted subsidies, tax credits, and public investment programs designed to support the growth of clean energy industries and accelerate decarbonization (Meckling 2021; 2025). Yet these policies increasingly incorporate a parallel objective: ensuring that domestic, rather than foreign, firms capture the economic benefits generated by the clean energy transition. The Inflation Reduction Act (IRA) in the United States is among the most consequential examples of this approach, combining unprecedented support for clean energy investment with provisions designed to channel the resulting economic gains toward domestic producers. This strategy reflects a growing belief that climate policy is more likely to endure when its economic benefits are concentrated among politically salient domestic constituencies (Meckling et al. 2015).

This combination of market expansion and preferential support for domestic firms creates a distinct tension in the design of contemporary green industrial policy: if domestic producers gain from these policies does it reflect growth in global markets or simply a reallocation of productive activities across national borders? We develop a framework that distinguishes between market expansion, in which industrial policy enlarges global markets for clean technologies, and market diversion, in which it redirects existing production and market share toward domestic firms. Whether the international effects of green industrial policy are dominated by market expansion or market diversion is theoretically ambiguous. Yet the answer to this questions is increasingly important as governments rely more heavily on industrial policy in hopes of accelerating clean energy transition.

We examine this question in the context of the 2024 U.S. presidential election, which provided a negative signal regarding the future durability of U.S. climate policy, particularly the 2022 IRA. The IRA represents the largest expansion of U.S. support for clean energy industries in history, providing hundreds of billions of dollars in subsidies and tax incentives for renewable energy, batteries, electric vehicles, and related technologies

([Gazmararian, Jensen and Tingley 2026](#)). At the same time, many of these incentives are coupled with preferential provisions, including domestic content requirements and other measures designed to direct the gains from market expansion toward U.S. firms ([Leonelli and Clora 2024](#)). These features—along with the scale of the IRA’s proposed investments—make it an especially salient case for comparing the policy’s effects on the expansion of global green markets versus redistribution of international competitive advantage.¹

The 2024 U.S. presidential election provides a valuable opportunity to examine how investors perceived the distribution of benefits generated by the IRA. Unlike previous elections, in which climate policy featured primarily as a background or symbolic issue, Donald Trump’s explicit promise to repeal the IRA occupied a central place in his 2024 campaign. Throughout the campaign, Trump pledged to repeal or weaken major provisions of the IRA—which he derided as the “Green New Scam”—and to curtail federal support for clean energy industries ([Trump Campaign 2024](#); [Colgan and Genovese 2025](#)). His electoral victory therefore provided investors with a clear signal that the probability of future policy retrenchment had increased ([Martins et al. 2025](#)). Examining how these revised expectations affected the valuation of clean energy firms provides insight into which mechanism—market expansion or market diversion—investors expected to dominate the IRA’s international effects.

We employ this negative shock to the perceived durability of the IRA as a means of testing which of our competing mechanisms dominate investor perceptions. If investors viewed the IRA primarily as expanding global markets for clean technologies, then reducing the expected durability of the legislation should lower the expected value of the global green sector by reducing future demand, investment, and innovation. Under the

¹Recent work by international organizations has highlighted the international competitive effects of industrial policy, including its implications for trade, investment, and competitiveness across countries ([Hodge 2024](#); [Huang et al. 2025](#)). Our contribution is to distinguish between market-expansion and market-diversion channels within green industrial policy and to examine which investors expect to predominate.

market expansion hypothesis, the aggregate effect of Trump’s election on global green firms should therefore be negative. By contrast, if investors viewed the IRA primarily as redirecting pre-existing production, investment, and market share toward U.S. firms, then weakening the legislation should reduce the expected value of U.S. firms while benefiting foreign competitors. Under the market diversion hypothesis, Trump’s election should therefore generate negative abnormal returns for U.S. firms but positive abnormal returns for foreign competitors, particularly Chinese firms ([Mazzocco 2022](#); [Meckling 2025](#)).

We estimate an event study of publicly traded clean energy firms in the United States, China, and Germany. We find no evidence of a substantial aggregate effect of anticipated IRA retrenchment across the global clean energy sector. Estimated abnormal returns are economically small and precisely estimated around zero, providing little support for the market expansion hypothesis. Instead, we find pronounced cross-border heterogeneity. U.S. clean energy firms experienced large negative abnormal returns, whereas Chinese firms experienced positive abnormal returns. German firms exhibited smaller and generally negative responses. The differences between U.S. and Chinese firms are substantively large and precisely estimated, and are driven primarily by firms producing solar panels and other infrastructural components of clean energy systems. Taken together, these findings suggest that investors expected the IRA’s market diversion effects to outweigh its contribution to expanding global clean energy markets.

Our analysis contributes to the growing political economy literature on green industrial policy by shifting attention from its domestic foundations to its international consequences. Existing work has emphasized how industrial policy accelerates decarbonization and its ability to build domestic political support ([Gazmararian, Jensen and Tingley 2026](#); [Meckling 2021](#)). In contrast, literature on the international effects of climate policy has remained largely agnostic about the specific instruments by which climate policy is implemented ([Genovese 2019](#); [Cory, Lerner and Osgood 2021](#); [Colgan, Green and Hale 2021](#)). We bridge these two literatures, arguing that modern green industrial policies

simultaneously expand markets and reshape the international distribution of industrial activity. Distinguishing between these two channels helps to clarify why the international consequences of green industrial policy have proven difficult to anticipate ([Melnick 2026](#)).

More broadly, the results highlight a potential tension embedded in the political logic of green industrial policy. Existing scholarship emphasizes the potential of green industrial policy to cultivate domestic political constituencies that favor pro-climate reform ([Meckling et al. 2015](#); [Meckling 2021](#)). This builds on a growing emphasis within the field on the importance of the domestic distributional foundations of successful climate policy ([Aklin and Mildenberger 2020](#), though see also [Kennard and Schnakenberg \(2023\)](#) and [Rowan \(2025\)](#)). However, measures intended to strengthen domestic clean energy industries may do so by undermining the profitability of clean energy firms abroad. Strategies designed to build durable domestic constituencies for climate action may therefore not reinforce, and may even undermine, the economic interests that could support decarbonization in other jurisdictions.

The International Effects of Green Industrial Policy

Green industrial policy has become an increasingly central feature of national climate strategies. Although governments have long used industrial policy to shape economic development, what distinguishes the current moment is the scale, ambition, and political prominence with which these tools are being deployed in pursuit of decarbonization. Green industrial policy relies on state intervention to direct economic activity toward the development and eventual deployment of clean technologies through instruments such as subsidies, tax credits, and concessional finance. This transformation is evident in the proliferation of major policy initiatives, from the U.S. Inflation Reduction Act and the European Union’s Green Deal Industrial Plan to comparable programs in China, Japan, South Korea, and across the Global South ([Meckling 2021](#); [Rodrik 2014](#)).

This resurgence of green industrial policy reflects both economic and political considerations. Economically, industrial policy is often justified as a response to market failures that impede investment in emerging clean technologies. Politically, however, it offers governments a means of building durable domestic support for climate action by creating jobs, attracting investment, and cultivating industrial constituencies with a stake in the energy transition. Unlike traditional climate instruments, which primarily seek to internalize the social costs of greenhouse gas emissions, green industrial policy also aims to generate visible economic benefits that strengthen political support for continued decarbonization ([Stern and Stiglitz 2023](#); [Meckling, Sterner and Wagner 2017](#)).

The central aim of green industrial policy therefore is to expand markets for low-carbon technologies. In the absence of state support, firms and consumers may underinvest in these technologies because of coordination failures, financing constraints, or a failure to internalize the long-run social benefits of early investment. By reducing barriers to investment, industrial policy accelerates the development, production, and deployment of clean technologies while stimulating additional private investment and innovation. As production expands, learning-by-doing and economies of scale reduce costs, encouraging broader adoption and reinforcing market growth. The result is an expansion in the scale of clean technology markets, supporting the diffusion of low-carbon alternatives throughout the economy ([Rodrik 2014](#)).

A classic illustration is the global solar photovoltaic industry during the 2000s. Germany’s generous feed-in tariffs created a large and predictable market for solar panels, helping to accelerate deployment, reduce costs, and stimulate investment throughout the global supply chain. Yet many of the resulting production and learning effects occurred outside of Germany. Chinese manufacturers rapidly expanded capacity, benefiting from scale economies and learning-by-doing, and soon emerged as dominant global producers. The German case therefore illustrates both the promise and the political vulnerability of market-expanding green industrial policy: demand-side support can accelerate global

technological change, but the industrial gains from that expansion may accrue elsewhere (Nemet 2019; Nahm 2021).

This experience helps explain why contemporary green industrial policies have over time become increasingly preferential in their design. Rather than simply accelerating the deployment of clean technologies, governments increasingly seek to shape where those technologies are produced and how global green supply chains are organized. Recent policies therefore extend beyond technology-neutral support for decarbonization to include provisions explicitly intended to encourage domestic production and investment. These measures include domestic content requirements, local manufacturing incentives, production subsidies tied to domestic facilities, and tax credits contingent on local sourcing or assembly. As a result, contemporary green industrial policy increasingly combines broad support for clean technologies with targeted incentives designed to influence the location of production (Meckling 2025; Mazzocco 2022).

The growing use of preferential provisions therefore creates a tension in the design of contemporary green industrial policy. On the one hand, industrial policy seeks to accelerate investment and expand global markets for clean technologies by overcoming market failures. On the other hand, preferential provisions may generate beggar-thy-neighbor effects by diverting production, investment, and market share toward domestic firms at the expense of foreign competitors. We refer to these competing channels as market expansion and market diversion, respectively.

This framework parallels the well-established distinction between business expansion and business stealing in the industrial organization literature (Mankiw and Whinston 1986; Tirole 1988), and also bears some resemblance to the classic distinction between trade creation and trade diversion in the international trade literature (Viner 1950). Firms can increase profits either by expanding the overall size of a market or by capturing market share from competing firms. An analogous distinction applies to green industrial policy. Rather than maximizing firm profits, governments seek to increase domestic investment,

employment, and industrial capacity. They can do so either by expanding the overall market for clean technologies or by redirecting production and market share toward domestic firms. The central question is not simply whether green industrial policy benefits domestic producers, but whether it does so primarily by enlarging global markets or by redistributing pre-existing economic activity across countries.

While both market expansion and market diversion are likely to coexist in many green industrial initiatives, it is an open question which is likely to dominate in any particular empirical instance. This question is theoretically ambiguous and, importantly, cannot be inferred from policy design alone: provisions directed towards both objectives coexist in most policy contexts. What determines the ultimate impacts of the legislation depend on both national economic circumstances and the reactions of firms, investors, and consumers ([Meckling 2025](#); [Mazzocco 2022](#)). Distinguishing which of these competing effects is likely to dominate therefore requires an empirical evaluation of their relative impacts.

Empirical Setting and Predictions

We test competing implications of market expansion versus market diversion in the context of the 2022 U.S. Inflation Reduction Act. The IRA represents one of the largest green industrial policy initiatives enacted by any advanced economy. The legislation provides an estimated \$370 billion in federal support for clean energy, transportation, manufacturing, and related technologies through a combination of tax credits, grants, and other incentives. Subsequent analyses suggest that the legislation has already catalyzed hundreds of billions of dollars in additional private investment, reflecting its central role in shaping expectations about the future direction of U.S. climate and industrial policy. Given both the scale of the legislation and the size of the U.S. economy, the IRA has the potential to influence investment decisions, supply chains, and competitive dynamics well beyond U.S. borders ([International Energy Agency 2024](#); [Bermel et al. 2024](#)).

Beyond considerations of scale, the IRA exemplifies the broader resurgence of green industrial policy described above. Rather than relying primarily on carbon pricing or environmental regulation, the legislation seeks to accelerate decarbonization through production and investment incentives for clean technologies. It therefore places industrial policy and public investment at the center of the U.S. climate strategy while seeking to stimulate private investment in strategically important industries. As a result, the IRA has become a leading example of the renewed role of industrial policy in climate governance and a focal point in debates over the state’s role in shaping the energy transition ([Bistline and Wolfram 2025](#); [Leonelli and Clora 2024](#)).

The IRA also exemplifies the growing use of preferential provisions within contemporary green industrial policy. While the legislation seeks to accelerate the deployment of clean technologies, many of its benefits are explicitly tied to domestic production, manufacturing, or sourcing. Tax credits for electric vehicles, batteries, and renewable energy technologies incorporate domestic content requirements or provide additional incentives for goods produced in the United States. These provisions reflect the Biden administration’s broader objectives of rebuilding domestic manufacturing, creating high-quality employment, and strengthening strategically important industries alongside decarbonization ([Leonelli and Clora 2024](#)).

These features make the IRA an ideal case for evaluating the framework developed above. The legislation simultaneously embodies both the logic of market expansion and market diversion, making its overall international consequences difficult to infer from its design alone. Whether investors expect the IRA primarily to expand global markets for clean technologies or to redirect production and market share toward U.S. firms is therefore an empirical question. We address this question by examining how changes in the expected durability of the legislation affect investor valuations of green firms in the United States and abroad.

Donald Trump’s 2024 election victory provides a uniquely informative setting in which

to evaluate these competing channels. Unlike many elections, in which the implications for climate policy are uncertain or only indirectly related to specific legislation, the 2024 presidential campaign featured explicit and sustained debate over the future of the Inflation Reduction Act. Throughout the campaign, Trump repeatedly criticized the legislation, describing it as economically wasteful and pledging to repeal or substantially scale back many of its provisions if elected.² His victory therefore constituted a negative shock to expectations about the future durability of the IRA. Observing how investors updated their valuations in response to this shock provides evidence about the effects they expected the legislation itself to have ([Trump Campaign 2024](#); [Friedman 2024](#)).

Our framework yields contrasting empirical predictions. If investors viewed the IRA primarily as expanding global markets for clean technologies, then reducing the expected durability of the IRA should lower the expected value of the global green sector by reducing future demand, investment, and innovation. Under the logic of market expansion, the aggregate effect of Trump’s election on global green firms should be negative. By contrast, if investors viewed the IRA primarily as redirecting production, investment, and market share toward U.S. firms—market diversion—then weakening the legislation should reduce the expected value of U.S. firms while benefiting foreign competitors. Under this logic, Trump’s election should generate negative abnormal returns for U.S. firms but positive abnormal returns for foreign-based competitors, particularly Chinese firms, which are widely recognized as major competitors in global clean technology markets ([Mazzocco 2022](#)).

²Trump famously derided the IRA as the “Green New Scam” and complained about its impacts on electricity reliability, the economy, and other outcomes ([Friedman 2024](#)).

Research Design

We examine how the 2024 U.S. presidential election reshaped investor expectations regarding the Inflation Reduction Act using an event study of publicly traded clean energy firms headquartered in the United States, China, and Germany. The election provided investors with new information about the expected durability of the IRA, and post-election stock prices therefore reflect updated expectations about the policy’s economic consequences. Comparing abnormal returns across countries enables us to evaluate whether investors expected anticipated IRA retrenchment primarily to reduce sector-wide demand for clean energy technologies or to redistribute competitive advantage across firms operating in different national jurisdictions.³

Data. We collect daily stock price data from Compustat Global and Compustat North America via Wharton Research Data Services, covering the six months prior to the election through December 31, 2024. Our sample includes publicly traded clean energy firms listed on the New York Stock Exchange and NASDAQ in the United States, the Shanghai and Shenzhen Stock Exchanges in China, and the Frankfurt Stock Exchange in Germany.

To identify firms exposed to climate and industrial policy, we use two complementary classification systems. We begin by identifying firms classified under Bloomberg Industry Classification System renewable energy categories and then supplement this list with firms identified through the LSEG/Refinitiv Thomson Reuters Business Classification system. The resulting datasets are merged and deduplicated to produce a comprehensive sample of firms operating in clean energy manufacturing and related supply chains across the United States, China, and Germany.

We organize firms into four sectors: solar, wind, infrastructure, and electric vehicles.

³Event studies are widely used in political science and economics to examine how investors respond to major political developments and incorporate new information into asset prices. In the context of climate politics, see [Aklin \(2018\)](#); [Bayer, Crippa and Genovese \(2025\)](#); [Martins et al. \(2025\)](#); [Voeten \(2025\)](#).

The solar sector encompasses the photovoltaic supply chain, including firms producing polysilicon, wafers, modules, inverters, and related equipment. The wind sector includes turbine manufacturers and component suppliers. The infrastructure sector covers enabling technologies such as energy storage, grid equipment, fuel cells, and related clean energy supply chain activities. We combine several infrastructure-related subsectors because individually they contain too few publicly listed firms, particularly in Germany, to support reliable estimation. Finally, the electric vehicle sector includes EV manufacturers and EV-focused supply chain firms. Sector assignments are based on Bloomberg and LSEG classifications and were harmonized across the two datasets.

Our objective is to capture firms whose profitability is directly tied to the manufacture and deployment of clean energy technologies. We therefore exclude fossil fuel firms, electricity generators and project operators, and firms for which renewable energy activities are incidental to their broader business operations. We additionally excluded six firms based on a manual review of their business descriptions. Details on sector definitions, firm-level exclusions, and coding decisions are reported in Appendix A2. The final sample comprises 147 firms: 85 in China, 13 in Germany, and 49 in the United States.

Estimation. We employ an event study design to estimate how the 2024 U.S. presidential election affected expectations regarding the future profitability of green firms. Event studies are widely used in finance and political economy because stock prices incorporate available information about expected future earnings (e.g., [Bayer, Crippa and Genovese 2025](#); [Voeten 2025](#); [Martins et al. 2025](#)). Abnormal stock returns therefore provide a measure of how market participants update expectations following politically consequential events.

We model each firm’s daily log returns, computed as first differences of log closing prices, using the Fama–French three-factor model ([Fama and French 1993](#)). For firm i in country c on day t , the estimating equation is:

$$R_{it} = \beta_0 + \beta_1 RM_{ct} + \beta_2 SMB_{ct} + \beta_3 HML_{ct} + \varepsilon_{it}.$$

Here, RM_{ct} denotes the market return, SMB_{ct} the size premium, and HML_{ct} the value premium. We use country-specific factor series (U.S., Europe, and emerging markets) from Kenneth French’s data library to account for systematic differences in risk across contexts. American and European factors are available daily. Emerging market factors, used for Chinese firms, are available monthly; we assign monthly values to each trading day within the corresponding month as is standard.

The baseline estimation window spans 180 calendar days prior to the election, excluding the five days immediately preceding it. We estimate firm-specific factor models over this window and use them to generate predicted returns during the post-election event window. Abnormal returns are the difference between realized and predicted returns.

Our primary outcome is cumulative abnormal returns (CARs) over the two trading days following the election. We focus on a two-day event window because election outcomes are revealed on a rolling basis throughout election day, with many results finalized only after markets close. This implies that the full informational content of the result is incorporated into prices on the subsequent trading day. This choice is standard in event studies with staggered information release ([MacKinlay 1997](#); [Kothari and Warner 2007](#)).

We aggregate firm-level CARs using inverse-variance-weighted (IVW) regression, which places less weight on firms with imprecisely estimated abnormal returns. Our baseline specification estimates average abnormal returns by country, and we additionally report results by country–sector combinations. All estimates use heteroskedasticity-robust standard errors.

Anticipation Effects. A potential concern is that pre-election polling and prediction markets may have led investors to partially anticipate the outcome, attenuating observed

price movements on election day. Trump’s win probability rose substantially in October 2024, suggesting that some information may have been incorporated into prices in advance. We address this in two ways. First, we exclude the five days prior to the election from the estimation window to avoid contaminating pre-treatment return estimates. Second, we present a robustness specification in the appendix excluding the 20 days prior to the election, covering the period of most rapid updating in prediction markets. Results are stable across these specifications.

Robustness. We conduct a series of robustness checks, reported in the Appendix (Figures A-1 through A-6 and Tables A-1 through A-11). Our baseline specification uses a 180-day estimation window, excluding the five days immediately prior to the election. We also estimate models using shorter windows of sixty and thirty days, as well as a specification excluding the twenty days prior to the election to address potential anticipation effects. Shorter estimation windows increase the variance of the Chinese estimates in particular, as Fama-French emerging market factors are only available monthly. We additionally estimate cumulative abnormal returns over three-day and five-day event windows. Results are substantively similar across specifications.

Along with our preferred Fama-French factor specification, we estimate a standard market model using country-specific indices (S&P 500, DAX 40, and CSI 300). Finally, we implement placebo tests by assigning pseudo-event dates thirty and sixty days prior to the election and re-estimating the full model using non-overlapping estimation windows. Across all specifications, results are stable and directionally consistent with the main findings.

Our analysis proceeds in two stages. We first estimate average abnormal returns for firms in each country. We then examine heterogeneity across sectors and firms to assess whether responses differed across segments of the green economy. Together, these analyses provide evidence on whether anticipated changes in U.S. climate policy altered

the valuation of firms operating both within and beyond the United States. Our objective is not to estimate the realized effects of future policy changes, but rather to examine how investors updated assessments of firm value following new information regarding the future durability of the IRA.

Results

Figure 1 presents the main results. The left panel displays mean cumulative abnormal returns (CARs) over the two days immediately following the election for green firms in each of the three countries and across the pooled sample; the right panel reports the corresponding pairwise differences relative to the United States.

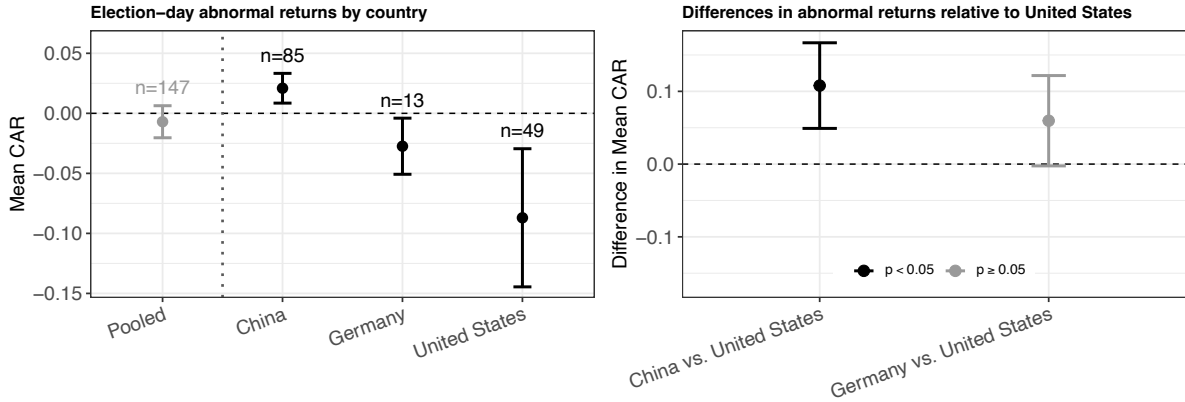


Figure 1: Impacts of 2024 Presidential Election on Green Firms

Note: Mean country-level abnormal returns with 95% confidence intervals using inverse-variance-weighted regression and heteroskedasticity-robust standard errors. Firm-level abnormal returns obtained with a three-factor Fama-French model.

The pooled estimate across all firms is small and statistically indistinguishable from zero. Although the analysis does not establish that the aggregate effect was exactly zero, the confidence interval indicates that any sector-wide impact was likely to have been modest. Investors therefore do not appear to have expected Trump’s election to substantially increase or decrease the aggregate value of the clean energy sector predicted under the hy-

pothesis of market expansion. This aggregate pattern, however, masks striking differences across countries.

U.S. clean energy firms experienced a large and statistically significant decline in market valuation following Trump’s election. Across U.S. firms in our sample, the estimated two-day cumulative abnormal return (CAR) is -8.7 percentage points. By contrast, Chinese firms experienced positive abnormal returns, with a mean CAR of 2.1 percentage points. The estimated difference between Chinese and U.S. firms is 10.8 percentage points, which is both economically large and precisely estimated. German firms also experienced negative abnormal returns, with an estimated mean CAR of -2.7 percentage points, although the difference relative to U.S. firms is not statistically significant at conventional levels. Given the relatively small number of German firms in the sample, these estimates are less precise than those of either the United States or China.

Taken together, these findings point to a common interpretation. Investors appear to have expected the principal consequence of IRA retrenchment to be a redistribution of competitive advantage across firms operating in different national jurisdictions rather than a substantial contraction of clean energy markets. In other words, investors expect the IRA’s effects to operate predominantly via market diversion rather than via market expansion.

Figure 2 disaggregates these patterns by sector: solar, wind, infrastructure, and electric vehicles. As the number of observations declines at this level of disaggregation, many estimates are imprecise. Nonetheless, the directional patterns are broadly consistent with the country-level results and suggest that the market response was not confined to a single segment of the green economy.

The China–U.S. gap is largest and most precisely estimated in the solar sector, where Chinese firms outperform U.S. firms by 14.0 percentage points. Given China’s dominant position in photovoltaic manufacturing and the importance of U.S. policy support for solar deployment, this result suggests that investors expected changes in U.S. climate policy

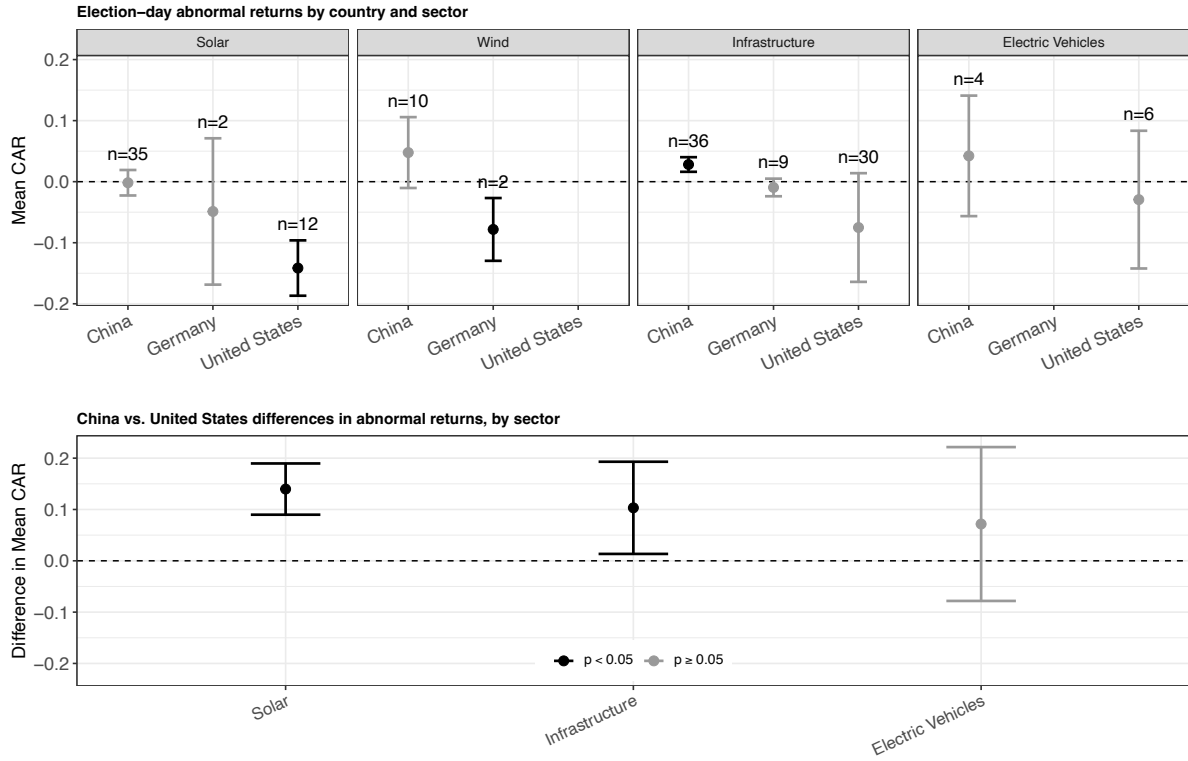


Figure 2: Cumulative Abnormal Returns by Country-Sector

Note: Mean country-sector-level abnormal returns with 95% confidence intervals using inverse-variance-weighted regression and heteroskedasticity-robust standard errors. Firm-level abnormal returns obtained with a three-factor Fama-French model.

to have particularly large consequences in globally integrated solar markets. A similarly large gap of 10.3 percentage points appears in the infrastructure sector. Estimates for electric vehicles are positive but imprecise, likely reflecting limited sample size. Notably, the largest differences emerge in sectors that have been central targets of industrial policy in both countries, indicating that investors expected anticipated changes in U.S. support for clean energy industries to affect firms operating throughout these markets.

Figure 3 presents firm-level CARs ranked within each country. The distribution of returns reinforces the aggregate results while highlighting within-country variation. Among U.S. firms, the distribution is strongly negatively skewed: most firms experience substantial losses, with some recording CARs between -30 and -50 percentage points. Only a small number of firms, primarily in infrastructure, show positive returns. Overall, losses

were widespread across the U.S. green sector rather than concentrated among a small number of firms, suggesting that the election altered expectations across a broad range of clean energy industries.

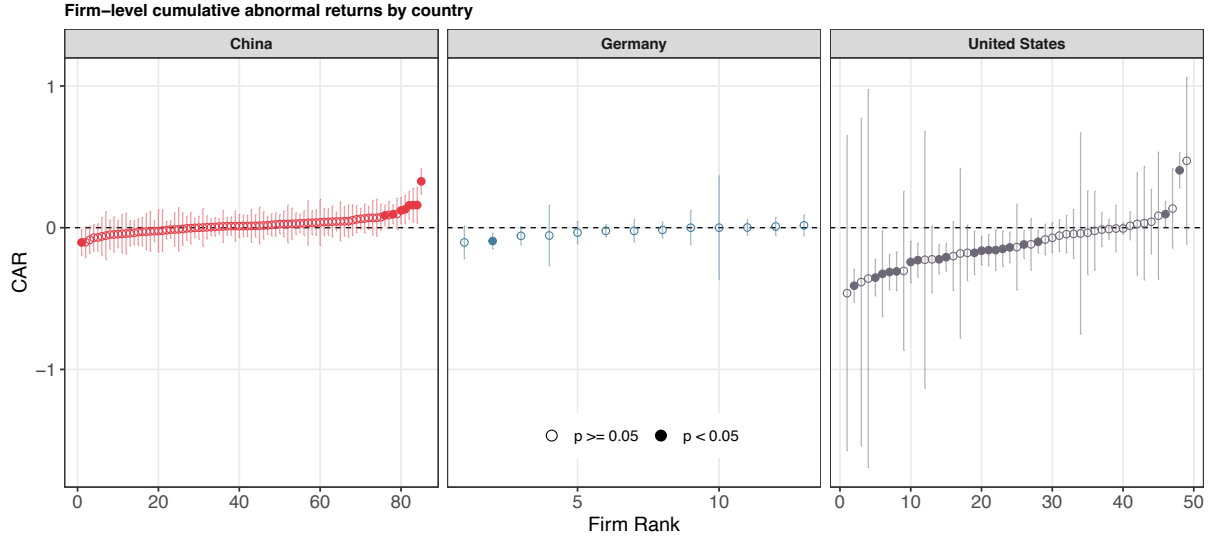


Figure 3: Firm-Level Abnormal Returns

Note: Firm-level abnormal returns with 95% confidence intervals obtained with a three-factor Fama-French model.

Among Chinese firms, the distribution is centered near zero with a pronounced right tail, including firms with gains of 10 to 30 percentage points. This suggests that the positive average effect is not driven by a small number of outliers. Approximately two-thirds of Chinese firms experienced positive abnormal returns, indicating that the observed cross-border differences reflect broad-based changes in valuation rather than the performance of a handful of firms.

The sector- and firm-level evidence points to a systematic reassessment of firm valuations following the election. The strongest responses appear in industries that have been central targets of green industrial policy, while the firm-level distributions indicate that these effects were widespread within countries. The concentration of the largest cross-national differences in solar and infrastructure suggests that investors expected the effects of U.S. climate policy retrenchment to be transmitted through specific clean energy

markets rather than affecting all firms uniformly. These patterns are consistent with the view that policy changes in one country can generate substantial valuation impacts for firms operating elsewhere in globally integrated industries.

Discussion

Our findings suggest that investors viewed the principal consequences of green industrial policy as redistributive. Following Trump’s election, U.S. clean energy firms experienced substantial losses in market value, while Chinese firms operating in similar sectors experienced gains. If investors had viewed the Inflation Reduction Act primarily as expanding global demand for clean energy technologies, a growing likelihood of policy retrenchment would be expected to reduce valuations across clean energy firms across the board. Instead, the divergent responses indicate that investors expected IRA retrenchment to reshape the relative competitive positions of firms operating in global clean energy markets. The prospect of policy retrenchment reduced the expected advantages enjoyed by U.S. firms while improving the outlook for foreign competitors. These findings suggest that preferential green industrial policies may strengthen domestic industries while simultaneously weakening the competitive position of clean energy firms abroad.

The cross-country heterogeneity also provides insight into the mechanism underlying investors’ expectations. Chinese firms experienced positive abnormal returns following Trump’s election, whereas U.S. firms experienced substantial losses. Chinese firms are major producers of solar panels, batteries, and other technologies that compete directly with U.S. manufacturers targeted by the IRA’s subsidies and domestic-content provisions (Nahm 2021; Allan and Nahm 2025). German firms exhibited smaller and generally negative responses, although these estimates are less precise. While our analysis does not identify the precise mechanisms driving these differences, the results are consistent with investors distinguishing between firms that compete directly with U.S. producers

and those occupying different positions within global clean energy supply chains.

These findings also have implications for how scholars understand the political economy of decarbonization. Existing research emphasizes how climate policy reshapes domestic coalitions by creating beneficiaries who subsequently support more ambitious climate action (Meckling et al. 2015; Colgan, Green and Hale 2021). Our results suggest that this political logic may not aggregate beyond individual countries. If green industrial policies redistribute economic opportunities across countries, some of the gains enjoyed by domestic beneficiaries may come at the expense of foreign competitors. Strategies that strengthen domestic political support for climate policy may therefore weaken the economic interests that could otherwise support decarbonization abroad. As governments increasingly use industrial policy not only to accelerate the energy transition but also to cultivate national clean energy industries, the politics of decarbonization may become increasingly shaped by international competition over the location of production, investment, and profits. Understanding this emerging political economy requires attention not only to how climate policy expands clean energy markets, but also to how it redistributes competitive advantage across them.

Several limitations qualify our conclusions. First, abnormal returns capture investor expectations rather than realized outcomes, though the subsequent passage of the One Big Beautiful Bill Act—which curtailed support for solar, wind, and electric vehicles—suggests those expectations were largely accurate. Second, our sample includes only publicly traded firms, omitting private producers and unlisted segments of the green economy that may respond differently. Third, although we observe substantial heterogeneity across countries and firms, we lack detailed data on firms’ production networks, sales geography, and positions within global value chains. As a result, we cannot directly test whether the observed variation reflects differences in exposure to U.S. policy, competitive relationships with U.S. producers, or integration into complementary segments of clean energy markets.

Conclusion

We demonstrate that changes in expected U.S. green industrial policy had international reverberations. But rather than inducing a global chill in clean technology markets, these shifts were distributive in nature: Chinese firms gained market value while U.S. firms suffered. We therefore attribute these effects to the redistribution of competitive advantage internationally rather than a sector-wide contraction in demand for clean energy technologies following Donald Trump’s election victory. More broadly, our findings suggest that investors viewed the Inflation Reduction Act not only as a mechanism for expanding clean energy markets, but also as a policy that altered the relative competitive positions of firms operating within them.

These results underscore the importance of incorporating international spillovers into analyses of green industrial policy. Existing research has primarily focused on the domestic consequences of these policies, emphasizing their capacity to accelerate investment, create employment, and cultivate political constituencies supportive of decarbonization. Our findings suggest that these domestic benefits may be accompanied by important international redistributive effects. Policies designed to strengthen domestic clean energy industries can simultaneously weaken the competitive position of firms abroad, generating winners and losers across national borders. As governments increasingly rely on industrial policy to pursue climate objectives, understanding these cross-border consequences will become essential for explaining both economic and political outcomes.

More fundamentally, our findings suggest that the consequences of green industrial policy may extend beyond the domestic coalition-building processes emphasized in existing research. The positive response of Chinese firms to anticipated U.S. policy retrenchment indicates that investors expected changes in climate policy in one country to reshape economic opportunities in another. In other words, market participants appear to view green industrial policy not simply as a tool for expanding clean energy deployment, but also as

a mechanism that redistributes investment and profitability across national jurisdictions. If these expectations are correct, then the political consequences of industrial policy may likewise spill across borders.

A central argument in recent scholarship is that industrial policy can strengthen support for climate action by creating firms, workers, and communities with a material stake in the continued expansion of clean energy industries ([Meckling et al. 2015](#); [Breetz, Mildenberger and Stokes 2018](#); [Meckling 2021](#)). Yet if the economic gains generated by these policies are partly redistributive, then coalition-building in one country may reshape political incentives elsewhere. Governments whose firms lose market share or investment may respond with industrial policies of their own, while firms that benefit from foreign policy retrenchment may become advocates for more ambitious domestic support. In this sense, green industrial policy may generate political feedback loops that extend beyond national borders.

Whether these dynamics ultimately accelerate or impede decarbonization remains unclear. The patterns we document suggest that investors expect green industrial policy to influence the geographic distribution of investment and profits within the clean energy economy. One possibility is that this competition encourages governments to expand support for low-carbon technologies in an effort to attract investment and build domestic clean energy industries. Another is that it contributes to growing protectionism, fragmented supply chains, and competition over the location of green production rather than the pace of emissions reductions. Understanding how these international political feedbacks emerge from the distributive consequences of green industrial policy represents an important challenge for future research. As industrial policy becomes an increasingly central component of climate strategy, the future of decarbonization may depend not only on how governments support clean energy industries at home, but also on how those policies reshape incentives abroad.

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